

Neal Daftary

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EDUCATION

Institute of Technology, Nirma University

B.Tech in CSE (AI & ML) | CGPA: 7.77 | JEE Mains 2024: **97.62 percentile**

Ahmedabad, India

2024–2028

WORK & RESEARCH EXPERIENCE

Undergraduate Student Researcher

CSE Dept., Nirma University — ISRO-Funded Project

Jan '26 – Present

Ahmedabad, India (Hybrid)

- Building a CV pipeline on TMC-2/OHRC lunar imagery (detection + segmentation + morphometry) to feed lunar rover navigation and mission planning algorithms.

AI Software Engineering Intern

MZHubtech

Oct '25 – Dec '25

Hamburg, Germany (Remote)

- Shipped a Next.js web-app on Azure (+**57% performance**, +**40% SEO**) via SSR, Core Web Vitals, bundle optimisation, and tree shaking; R&D on OCR, agentic AI, and CX chatbots shaped the product roadmap.

AI Intern

8xSports

Jun '25 – Sept '25

Bhilwara, Rajasthan, India (Remote)

- Built a **real-time visual jersey search engine** (645ms): YOLOv8 + DINOv2 384-dim features + LBP texture + FAISS across 301 profiles; earned a **Letter of Recommendation**.
- IEEE Sensor Letters** (SCI Q3, IF 2.2, Jan '26): CatBoost for fault detection in industrial robotic arms — **97.20% precision, F1 0.9718**, outperforming SVM and RF baselines. 🏆

PROJECTS

SOLV.ai — AI-Powered Voice Complaint Management System 🏆

Python, FastAPI, ONNX Runtime, DistilBERT-MNLI, MiniLM-L6, VADER, Twilio, Groq, Ollama

- NLP + voice stack for FMCG complaint triage: **dual-model ONNX ensemble** (DistilBERT-MNLI + MiniLM-L6) + VADER; **6-state FSM** with 5 agents, dual LLM (Groq/Ollama) + dual TTS fallback; **65% latency reduction** (35ms→12ms), **100% category accuracy**; \$1.83/M vs \$1,500 GPT-3.5.

Production RAG Chatbot — IAT Networks 🏆

🔴 Live

Python, FastAPI, ChromaDB, MiniLM-L6-v2, Groq (Qwen3-32B), SlowAPI, Docker, Railway

- Production RAG chatbot** for an ISP: query expansion → MiniLM → ChromaDB (top-8) → reranking (top-4) → Groq SSE; **4-layer GuardRail** (injection regex, PII, domain filter); Dockerized on Railway; **0% injection bypass**, **<800ms TTFT**, **zero downtime**.

MemoryLens — LLM Memory Decay Evaluation Framework 🏆

Python, Groq, sentence-transformers, NumPy, Streamlit, Plotly, GitHub Actions

- Open-source benchmark for LLM memory decay across long conversations: **3 backends** (Naive, RAG, Cascading Temporal) × **5 content-based metrics**; Cascading Temporal achieves **5.45× recall/token** vs naive at T=100 with **78% lower cost**; Streamlit dashboard; CI via GitHub Actions.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL

CS Concepts: OS, DBMS, CA, DSA, OOP

Full-Stack: REST APIs, SSE, Docker, GitHub Actions, Azure App Services, Azure Cosmos DB, Railway

AI / LLM / RAG: RAG Pipelines, LLM Orchestration, Prompt Engineering, Guardrails, Agentic Workflows

ML: Supervised & Unsupervised Learning, XAI/SHAP, Hyperparameter Tuning, Anomaly Detection, ONNX

CV: CNN, ViT, Object Detection, Semantic & Instance Segmentation, Image & Video Inference

Frameworks & Tools: React, Next.js, FastAPI, PyTorch, LangChain, OpenCV, HuggingFace, Groq, FAISS, ChromaDB, MLflow, scikit-learn, Pandas, NumPy, Git

EXTRACURRICULARS & ACHIEVEMENTS

- Student Chairperson**, ACM ITNU (Sept '25 – Present): Leading 150+ member community; launched “Prompt to Prototype” and mentorship tracks across AI/ML, Data Science, Cybersecurity, and Web Dev.
- Track Winner & Top-5 National**, HACKaMINeD 2026 (2,200+ participants) — LUMIN.AI: explainable ML for solar inverter predictive maintenance. 🏆
- 4th National, 6th Mitsubishi Electric Cup 2026** — SpectraScan: 92.35% segmentation accuracy (DINOv2+FPN-UNet), 86% dimensional validation precision. 🏆
- Advanced Learning Algorithms**, DeepLearning.AI (Stanford) | Score: **99.77%** 🏆 | **LeetCode Rating:** 1507